c) By Defn. 1 from Sec-1.7, m can be even or m can be odd.
∴ ∃ K∈Z s.t. m=2K or m=2K+1.
⌈m/2⌉ = ⌈2K/2⌉ = ⌈K⌉ = K. n = K+ε₂, where ε₂ = 0.
⌈(2K+1)/2⌉ = ⌈K+1/2⌉ = K+1 = (K+½)+½.
n = (K+½)+½, here ε₂ = 1/2.
∴ ε₂ can either be 0 or 1/2. ∴ 0 ≤ ε₁/2+ε₂ < 1
x/4 + ε₁/2 + ε₂ = n ⇒ ⌈x/4⌉ = n
∴ ⌈⌈x/2⌉/2⌉ = ⌈x/4⌉ for ∀x∈R ∎

d) ⌊√⌈x⌉⌋ = ⌊√x⌋ for all positive real numbers x.
Soln: Let, x = 8.1 ∴ ⌊√⌈x⌉⌋ = ⌊√⌈8.1⌉⌋ = ⌊√9⌋ = ⌊3⌋ = 3
⌊√8.1⌋ = ⌊2.84⌋ = 2. ∴ ⌊√⌈x⌉⌋ ≠ ⌊√x⌋ for all x∈R. ∎

e) ⌊x⌋+⌊y⌋+⌊x+y⌋ ≥ ⌊2x⌋+⌊2y⌋ for all real numbers x and y.
Soln: Let, x = 1.1 and y = 1.2
⌊x⌋+⌊y⌋+⌊x+y⌋ = ⌊1.1⌋+⌊1.2⌋+⌊1.1+1.2⌋
= 1+1+2 = 4.
⌊2x⌋+⌊2y⌋ = ⌊2×1.1⌋+⌊2×1.2⌋ = ⌊2.2⌋+⌊2.4⌋ = 2+2 = 4

7) For each of these partial functions, determine its domain, codomain, domain of definition, and the set of values for which it is undefined. Also, determine whether it is a total function.
a) f: Z → R, f(n) = 1/n
Soln: 0∈Z. f(0) = 1/0 which is undefined.
domain of defn: Z−{0} domain: Z. codomain: R.
Set of values for which it is undefined: {0}. Not a total function. ∎
b) f: Z → Z, f(n) = ⌈n/2⌉
Soln: domain of defn: Z, since ⌈n/2⌉ is defined for each Z.
domain = codomain = Z
The set of values for which it is undefined: ϕ. A total function. ∎